Aaryan S. Maralihalli

AI / ML Engineer — Fintech Systems

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Summary

Fintech-native AI engineer: six months **inside production identity-verification/KYC systems** for banks and lenders, plus hands-on LLM engineering — **RAG with RBAC and immutable audit logging**, **LoRA fine-tuning**, and real-time event pipelines. **IEEE-published** (interpretable ML, ROC-AUC 0.757). Builds AI the way payments infra demands it: measured, auditable, low-latency. CS grad, Jul 2026.

Experience

HyperVerge, Bangalore
Integration Engineer Intern Jan 2026 – Jun 2026
AI identity-verification / KYC platform for banks & fintechs

- Owned solutioning and integration of **production AI verification modules** — Video KYC, face match, liveness, name match, eSign, DigiLocker — for **6 enterprise fintech/lending clients** across **15+ lines of business**.
- Took **3 clients live at >95% verification conversion — a 10% lift**; drove post-go-live **business-impact monitoring and stabilization** (conversion, drop-off, edge-case failure analysis).
- Authored Solution Design Docs, issued production credentials, configured **API rate limits**, and debugged client integration failures in production; shipped a workaround for an eSign product gap to production.
- Built **"HV Deals Bot"** (Python + Slack) to track deal stages and document delivery process for the team.

Integrated Genetic Solutions, Bangalore
Project Intern, Medical AI Jun 2024 – Aug 2024

- Led a legacy-JS → **React (MERN)** migration improving processing of **5M+ records** on a HIPAA-compliant, compliance-heavy data product.

Selected Projects

SDIS — compliance-grade RAG service github.com/Aaryansm11/SDIS

- FastAPI document-intelligence platform with **JWT + RBAC, PII redaction (spaCy), immutable audit logging + digital signatures**, FAISS retrieval over PostgreSQL, rate limiting; Dockerized on AWS ECS; **pytest + GitHub Actions CI** — security and auditability as first-class design constraints.

Self-Supervised Cardiac Risk Model (IEEE-published) github.com/Aaryansm11/ecg-ppg-arrhythmia-prediction

- Contrastive learning (SimCLR/NT-Xent) on MIMIC-III biosignals (133K segments); custom 1D-CNN encoder + Random Forest; **ROC-AUC 0.757**, benchmarked vs clinical baseline; explainability via saliency + integrated gradients — rigorous, evaluated ML on noisy real-world signals.

Live Emostream — real-time event pipeline github.com/Aaryansm11/Live_emostream

- **Kafka + Spark + Redis**, concurrent broadcast with nginx load balancing; **load-tested to 1,000 concurrent users** — the shape of transaction-stream monitoring.

LANGTRON github.com/Aaryansm11/LANGTRON

- Fine-tuned **Mistral-7B (LoRA)**, dual-model architecture, **FAISS retrieval sub-100ms**, fully local.

Skills

- **ML / LLM:** PyTorch, HuggingFace Transformers, LoRA fine-tuning, RAG, FAISS, spaCy, BERT, contrastive learning, explainable AI
- **Fintech / Domain:** KYC & identity verification, onboarding funnels, compliance-aware design (RBAC, audit trails, PII handling), client solutioning
- **Systems:** Python, FastAPI, Kafka, Redis, Spark, PostgreSQL, Docker, AWS (EC2/S3/ECS), GitHub Actions CI, SQL, Java; DSA (150+ LeetCode)

Publication & Education

- **IEEE ICASI 2026 (Kyoto), Oral Presentation** — *"An Interpretable Model for Stroke Prediction and Arrhythmia Classification using ECG & PPG."*
- **B.Tech, Computer Science** — PES University, Bangalore — 2022 – Jul 2026